

Remark 4.3. The definition of the gcd in eq. (2) requires all p^n polynomials $f \in \mathbb{F}_p[T]$ of degree $< n$, which is not practical for computations. But a subset of these polynomials is usually sufficient. Let us explain this in more detail.

First we may assume that f is monic (since $p-1 \mid n-1$, constants $u \in \mathbb{F}_p$ satisfy $u^n = u$ anyway) and $\deg(f) > 0$. Choose an enumeration $f_1, f_2, f_3, \dots$ of all monic polynomials of degree $< n$ with $\deg(f_i) \leq \deg(f_{i+1})$. It is recommended to start with $f_1 = T$ and $f_2 = T+1$. Then the “partial gcd”

$$g_i := \gcd(f_1^n - f_1, \dots, f_i^n - f_i),$$

can be computed recursively via $g_{i+1} = \gcd(g_i, f_{i+1}^n - f_{i+1})$. If some g_i already satisfies $g_i \mid T^{p^k} - T$, we are done, even when maybe $g_i \neq g$. If not, compute g_{i+1} and continue. It turns out (see the examples below and Remark 4.17) that in many cases a small i will do the job, and that only a few, mostly linear polynomials are required for the $f_1, \dots, f_i$.

It would be very much desirable to give a “uniform” equational proof of Theorem 4.1, which thus can be written down without any specific choice of (n, p) and also without any tedious calculations. But it is not clear if it exists.

In any case, using a computer algebra system does help to find the equations. For the examples below, we have again used SageMath, the code can be found in appendix A.3.

Remark 4.4. Because of Remark 3.6, the polynomial g is not always equal to $T^{p^k} - T$, even though in the examples below it will often be the case. In general, one can show that g is the product of all monic irreducible polynomials in $\mathbb{F}_p[T]$ whose degree d satisfies $p^d - 1 \mid n - 1$. For $k = 1$ this means $g = T^p - T$.

We start with some examples where linear polynomials are sufficient for computing g .

Example 4.5. Consider $n = 3$ and $p = 2$. Then $k = 1$. The gcd of $T^3 - T$ and $(T+1)^3 - (T+1)$ in $\mathbb{F}_2[T]$ is $T^2 - T$ with the linear combination

$$T^2 - T = (T^3 - T) + ((T+1)^3 - (T+1)).$$

If A is any ring with $2 = 0$ and $x \in A$, the equation

$$x^2 - x = (x^3 - x) + ((x+1)^3 - (x+1))$$

can simply be verified by hand. This shows that any 3-ring with $2 = 0$ is a 2-ring. Here is a more compact version: We have $x+1 = (x+1)^3 = x^3 + 3x^2 + 3x + 1 = x + x^2 + x + 1$, hence $x^2 = x$. Alternatively, Lemma 2.15 gives directly $3x + 3x^2 = 0$, hence $x^2 = x$.

Example 4.6. Consider $n = 5$ and $p = 2$. Then $k = 1$ and the extended Euclidean algorithm yields

$$T^2 - T = (T+1) \cdot (T^5 - T) + T \cdot ((T+1)^5 - (T+1)) \in \mathbb{F}_2[T].$$

This produces the following equational proof that every 5-ring with $2 = 0$ is a 2-ring: We have $x+1 = (x+1)^5 = x^5 + x^4 + x + 1$, which gives $x^4 = x^5 = x$, hence $x = x^5 = x^2$. Alternatively, we may argue with Lemma 2.15 again and get $0 = 5x + 10x^2 + 10x^3 + 5x^4 = x + x^4$, so that $x^4 = x$ and then $x = x^2$.

Example 4.7. Consider $n = 5$ and $p = 3$. Then $k = 1$ and

$$T^3 - T = T \cdot (T^5 - T) - (T+1) \cdot ((T+1)^5 - (T+1)) \in \mathbb{F}_3[T].$$

This leads to the following equational proof that every 5-ring with $3 = 0$ is a 3-ring: We compute $x+1 = (x+1)^5 = x^5 + 2x^4 + x^3 + x^2 + 2x + 1 = 2x^4 + x^3 + x^2 + 1$, hence $x^4 = x^3 + x^2 - x$ (†). Multiplying with x and using $x^5 = x$ gives $x = x^4 + x^3 - x^2$. Plugging this back into (†) gives $x^4 = x^2$ and hence $x = x^5 = x^3$.

Example 4.8. Consider $n = 6$, hence $p = 2$ and $k = 1$. Then

$$T^2 - T = T^2 \cdot (T^6 - T) + (T^2 + 1) \cdot ((T+1)^6 - (T+1)) \in \mathbb{F}_2[T]$$

leads to the following equational proof that every 6-ring is a 2-ring: Since 6 is even, we have $-x = x$. Next, we compute $x + 1 = (x + 1)^6 = x^6 + x^4 + x^2 + 1 = x + x^4 + x^2 + 1$, hence $x^4 = x^2$. Thus, $x = x^6 = x^2x^4 = x^2x^2 = x^4 = x^2$. Conversely, every 2-ring is a 6-ring by Corollary 2.10.

Example 4.9. Consider $n = 9$ and $p = 5$. Then $k = 1$, and

$$T^5 - T = 2T \cdot (T^9 - T) - (2T + 2) \cdot ((T + 1)^9 - (T + 1)) \in \mathbb{F}_5[T]$$

can be used to give an equational proof that every 9-ring with $5 = 0$ is a 5-ring.

Example 4.10. Consider $n = 10$, hence $p = 2$ and $k = 2$. Then

$$T^4 - T = (T^2 + 1) \cdot (T^{10} - T) + T^2 \cdot ((T + 1)^{10} - (T + 1)) \in \mathbb{F}_2[T]$$

shows that every 10-ring is a 4-ring. Conversely, every 4-ring is a 10-ring by Corollary 2.10.

Example 4.11. Consider $n = 17$ and $p = 3$. Then $k = 2$ and

$$T^9 - T = T \cdot (T^{17} - T) - (T + 1) \cdot ((T + 1)^{17} - (T + 1)) \in \mathbb{F}_3[T]$$

shows that every 17-ring with $3 = 0$ is a 9-ring. It produces the following equational proof: Lemma 2.15 together with $\binom{17}{i} \equiv (-1)^i \pmod{3}$ for $0 \leq i \leq 8$ yields

$$-x^{16} + x^{15} - x^{14} + x^{13} - x^{12} + x^{11} - x^{10} + x^9 + x^8 - x^7 + x^6 - x^5 + x^4 - x^3 + x^2 - x = 0.$$

Multiplying this with x and using $x^{17} = x$ gives

$$+x^{16} - x^{15} + x^{14} - x^{13} + x^{12} - x^{11} + x^{10} + x^9 - x^8 + x^7 - x^6 + x^5 - x^4 + x^3 - x^2 - x = 0.$$

Adding these two equations gives $2x^9 - 2x = 0$, so that $x^9 = x$.

Example 4.12. Consider $n = 35$. Then $p = 3$ and $k = 1$. In $\mathbb{F}_3[T]$ we find that $T^3 - T$ is the gcd of $T^{35} - T$ and $(T + 1)^{35} - (T + 1)$, so that every 35-ring is a 3-ring. The converse also holds by Corollary 2.10.

Example 4.13. Consider $n = 7$ and $p = 2$. Then $k = 2$. Here, the gcd with linear polynomials does *not* suffice. But we are lucky with a quadratic one:

$$\begin{aligned} T^4 - T &= (T^8 + T^6 + T^5 + T^3 + 1) \cdot (T^7 - T) \\ &\quad + (T^8 + T^6 + T^5 + T^3 + 1) \cdot ((T + 1)^7 - (T + 1)) \\ &\quad + ((T^2 + T + 1)^7 - (T^2 + T + 1)) \end{aligned}$$

There is a simpler expression with a cubic polynomial:

$$T^4 - T = (T^{15} + T^{12} + T^3 + 1) \cdot (T^7 - T) + T \cdot ((T^3 + 1)^7 - (T^3 + 1))$$

We get the following equational proof that every 7-ring with $2 = 0$ is a 4-ring: We compute

$$\begin{aligned} x^3 + 1 &= (x^3 + 1)^7 \\ &= x^{21} + x^{18} + x^{15} + x^{12} + x^9 + x^6 + x^3 + 1 \\ &= x^3 + x^6 + x^3 + x^6 + x^3 + x^6 + x^3 + 1 \quad (\text{by (2.9)}) \\ &= x^6 + 1, \end{aligned}$$

so that $x^3 = x^6$. Multiplying this with x and using $x^7 = x$, we get $x^4 = x$.

Example 4.14. Consider $n = 13$ and $p = 2$. Then $k = 2$. Here, we also need linear and quadratic polynomials: In $\mathbb{F}_2[T]$, one finds that $T^4 - T$ is the gcd of $T^{13} - T$, $(T + 1)^{13} - (T + 1)$, $(T^2 + T + 1)^{13} - (T^2 + T + 1)$. This shows that every 13-ring with $2 = 0$ is a 4-ring.

Example 4.15. Every 31-ring with $2 = 0$ is a 16-ring (and vice versa). In this example even polynomials of degree 4 are required: $T^{16} - T$ is the gcd of the six polynomials $f^{31} - f \in \mathbb{F}_2[T]$, where f belongs to $\{T, T + 1, T^2 + T + 1, T^4 + T + 1, T^4 + T^3 + 1, T^4 + T^3 + T^2 + T + 1\}$.

In all the examples so far, the polynomial g was just $T^{p^k} - T$. But we already mentioned in Remark 4.4 that this does not have to be the case. Here is the smallest counterexample:

Example 4.16. Let $n = 22$, hence $p = 2$ and $k = 6$. The gcd of $T^{22} - T$, $(T^2 + T)^{22} - (T^2 + T)$ in $\mathbb{F}_2[T]$ is equal to $g' := T^{10} + T^9 + T^8 + T^3 + T^2 + T$, and this divides $T^{64} - T$. This yields an equational proof that every 22-ring with $2 = 0$ is a 64-ring. (But this is clear anyway because of $21 \mid 63$, and it is unclear if this helps to prove commutativity of 22-rings.) Incidentally, we have $g' = g$, because Remark 4.4 implies that $g = T(T+1)(T^2+T+1)(T^3+T+1)(T^3+T^2+1) = g'$. The complexity of 22-rings has been already noted by Morita [Mo78]. There is no $m < 22$ such that every 22-ring is an m -ring, as can be seen by considering $\mathbb{F}_4 \times \mathbb{F}_8$.

Remark 4.17. When $k = 1$, i.e. p is the only power p^d with $p^d - 1 \mid n - 1$, let us call n *simple* at p . The previous examples and numerical experiments suggest that then linear polynomials are often enough to compute g , i.e. that $T^p - T = \gcd((T + u)^n - (T + u) : u \in \mathbb{F}_p)$. Let us call n *nice* for p in this case, otherwise *unpleasant*. See appendix A.4 for the SageMath code to find unpleasant numbers. For $p = 2$, there are 5484 numbers $n \leq 10\,000$ that are simple at 2, and only 142 of these (that is 2.6%) are unpleasant, the smallest example being $n = 74$. For $p = 3$, all of the simple numbers $n \leq 100\,000$ are nice, and Peter Müller found that the smallest unpleasant number is $n = 1 + (3^{16} - 1)/32 = 1\,345\,211$. For $p = 5$, the smallest unpleasant number is $n = 1 + (5^8 - 1)/3 = 130\,209$. For other prime numbers, the behavior seems to be similar: most simple numbers are nice.

5. The simple case

In this section, we will show how to produce, for every fixed $n > 1$ with a special property (see below), an equational proof of the commutativity of n -rings. As before, we will not be able to write down a single equational proof that works for all n . The proofs are not “uniform” and have to be written down for each n individually. But the recipe is always the same.

Definition 5.1. Let us call $n \in \mathbb{N}$ *simple* if $n > 1$ and every prime power q with $q - 1 \mid n - 1$ is actually a prime number. The simple numbers ≤ 100 are (see appendix A.5 for the SageMath code to generate them)

2, 3, 5, 6, 11, 12, 14, 18, 20, 21, 23, 24, 26, 30, 35, 38, 39, 42, 44, 45, 47, 48, 51, 54,
56, 59, 60, 62, 66, 68, 69, 72, 74, 75, 77, 80, 83, 84, 86, 87, 90, 93, 95, 96, 98.

They form the OEIS sequence A366343². The natural density of the set of simple numbers (if it exists) is approximately

$$\lim_{N \rightarrow \infty} \frac{\#\{n \leq N : n \text{ is simple}\}}{N} \approx 0.462118,$$

meaning that almost every second number is simple. By Remark 3.1, a number n is simple precisely when every n -field is a prime field.

We start by giving a quick proof of Jacobson’s Theorem for simple numbers.

Theorem 5.2. *Let A be a ring such that for every $a \in A$ there is some simple number $n > 1$ such that $a^n = a$. Then every element of A is a $\mathbb{Z}$ -linear combination of idempotent elements. In particular, A is commutative.*

²<https://oeis.org/A366343>